
Süssmayr’s Heirs: A Composition-Invariant View of Generative Stylistic Completion

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Abstract

Stylistic completion, extending an unfinished work in the style of its creator, is among the oldest practical problems in Western music, recurring from Süssmayr’s 1791 completion of Mozart’s *Requiem* through to the symbolic infilling models, lyrics-to-song systems, and voice cloning tools released in 2025–2026. These systems are studied as separate technical domains despite a shared task structure. We propose a unified framework in which stylistic completers, human or learned, are operators on a musical action space, and a feature is **composition-invariant** under a composer’s stylistic prior iff it is preserved by every continuation in the prior’s support. The kernel of composition-invariants identifies what makes a continuation belong to a composer’s voice: the features distinguishing stylistic identity from interpretive choice. We connect the framework to existing musicological theories (Schenker, Lerdahl–Jackendoff, Lewin, Tymoczko, Kramer); propose a diagnostic battery computable on contemporary symbolic and audio outputs; apply two diagnostics in a three-way comparison on Mozart’s *Lacrimosa* (Mozart prompt, Süssmayr, Suno V5), calibrated against within-Mozart baselines on the K.626 Kyrie; and draw implications for evaluation, licensing, and co-creative tool design at a moment when these systems face simultaneous research, commercial, and legal scrutiny.

1. Introduction

In 1791, Franz Xaver Süssmayr completed Mozart’s *Requiem* from sketches, fragments, and remembered conversations with the dying composer. The resulting score has been performed continuously for over two centuries, yet musicology has never stopped asking: which gestures belong to

Mozart, and which to Süssmayr? This question has motivated multiple modern completions (by Eybler, Maunder, Levin, and Druce), each offering a different answer to what a “Mozartean” continuation would sound like.

By 2026, generative AI has industrialized the completion task. Suno and Udio, proprietary systems, generate full songs from text prompts. YuE (Yuan et al., 2025), built on LLaMA2, provides open-source lyrics-to-song long-form generation. Muse (Jiang et al., 2026) fine-tunes on 116k licensed songs synthesized by Suno V5 for style-controlled generation. DiffRhythm, LeVo, and ACE-Step offer alternative long-form architectures. On the symbolic side, the MIREX 2025 Symbolic Music Generation challenge formalizes the task: given a four-bar piano prompt and optional upbeat, generate twelve bars of musically coherent continuation, a benchmarked version of Süssmayr’s problem. MIDI-RWKV (Zhou-Zheng & Pasquier, 2025) provides personalizable, long-context, multi-track symbolic infilling.

Voice cloning extends the same problem to speech. Cartesia’s Sonic uses state-space model architectures (from the S4/Mamba lineage) for low-latency TTS and voice cloning. MiniMax Speech 2.5 (August 2025) performs voice cloning across 40+ languages. These systems perform stylistic completion at the utterance level: given a voice sample, generate speech that is indistinguishable from what that speaker would have said.

We propose a historical arc that unifies these developments:

1. **1791, Süssmayr:** Human operator inferring style from limited data (sketches, oral instructions). Operators reside in human cognition.
2. **1956, Illiac Suite:** Hiller and Isaacson at the University of Illinois program ILLIAC I to generate a four-movement string quartet based on Renaissance counterpoint rules (Hiller & Isaacson, 1959). Operators are explicitly encoded, a “white-box” precedent.
3. **2024–26, Suno/Udio/MiniMax/Cartesia:** Operators learned from data, implicit rather than explicit. Invariance can no longer be read from rules; it can only be diagnosed empirically.

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This arc frames the operator transition from explicit to implicit as the central technical shift. It also frames the current litigation landscape. UMG settled with Udio in October 2025, announcing a joint AI music platform for 2026. Warner settled with Suno in November 2025; Suno acquired Warner’s Songkick and committed to transitioning to licensed models. Sony’s suits against Suno (Massachusetts) and Udio (SDNY) remain active, with rulings expected summer 2026 that may set precedent for the industry. UMG, Concord, and ABKCO filed a \$3 billion publishing suit against Anthropic in January 2026. GEMA’s German action against Suno was set for judgment in June 2026.

These settlements and pending cases create an operational need that current licensing language does not meet: a shared technical vocabulary for which features of an artist’s style constitute that artist’s identity. Our framework offers a candidate. We define a property as *composition-invariant* under a composer’s style prior if it is preserved by any operator that produces a stylistically legitimate continuation. This partitions stylistic features into a *kernel* (identity-constituting invariants that any “in-style” completion must preserve) and a surrounding cloud of legitimate variation that different completers may realize differently. We do not claim the kernel is a legal definition of protectable style (Section 6).

Contributions. We make four contributions:

1. We construct a historical arc connecting Süssmayr (1791), the Illiac Suite (1956), and contemporary learned generators, showing that stylistic completion is a recurring problem whose technical substrate has shifted from explicit to implicit.
2. We provide a formal definition of composition-invariance that distinguishes identity-constituting features (the kernel) from stylistic choices.
3. We propose a diagnostic battery of musicology-grounded invariants (voice-leading parsimony, motivic transformation overlap, harmonic-rhythm regularity, and others) computable on any symbolic or audio generation, and apply two of them to a comparison on Mozart’s *Lacrimosa*: Mozart prompt, Süssmayr completion, and two independent Suno V5 generations, calibrated against within-Mozart variation from the same Requiem.
4. We discuss implications for evaluation, the 2026 copyright landscape, and the design of co-creative tools that resist stylistic homogenization.

2. Background

2.1. Stylistic Completion in Musicological Practice

Stylistic completion has been a recurring practical and scholarly problem in Western music for over two centuries. Mozart’s *Requiem* is the most studied case. Mozart’s autograph terminates approximately at the eighth measure of the *Lacrimosa*, with sketches of varying completeness extending into other movements. Joseph Eybler accepted the commission in 1791 and orchestrated several movements before withdrawing; Franz Xaver Süssmayr then took over and produced the score that has remained the standard performance text. Modern recompletions (Beyer 1971, Maunder 1988, Levin 1993, Druce) have engaged systematically with the question of which features of Süssmayr’s continuation are stylistically Mozartean and which are not. The persistence of this scholarly activity across two centuries is itself evidence that the relevant question admits multiple defensible answers, with each completer occupying a different point in the space of “Mozartean continuations.”

Mozart is not unique. Bach’s *Die Kunst der Fuge* breaks off in the unfinished *Contrapunctus XIV* at the entrance of the B-A-C-H subject, with completions proposed by Tovey, Moroney, and others. Schubert’s Eighth Symphony (“Unfinished”) has attracted both completion attempts and extensive argument about whether it should be left as found. Mahler’s Tenth was completed by Deryck Cooke in 1976. Bruckner’s Ninth has multiple completions; Anthony Payne’s 1998 elaboration of Elgar’s Third Symphony sketches is recognized as a substantial recompositional achievement.

Several observations recur across this literature. First, completers face a methodological choice between mimicking the original composer’s surface texture and developing the opening material into longer-range structures; the two strategies tend to fail in different ways. Second, multiple completers of the same work disagree systematically along certain axes (orchestrational choices, harmonic ambition) while converging on others (basic voice-leading propriety, formal proportion). Third, the existence of multiple defensible completions of one work gives the field something no single completion can supply: empirical access to which features of the original style require preservation across all valid continuations, and which features admit defensible variation. The standard musicological reading of Süssmayr, that his command of part-writing was idiomatic but his harmonic and orchestrational decisions reveal his hand (Wolff, 1994), is in effect a claim about which features lie inside Mozart’s stylistic kernel and which do not.

2.2. Computational Completion

The AI lineage runs in parallel and is much shorter. Hiller and Isaacson’s Illiac Suite (1956) was the first composi-

tion produced by a digital computer, generated by hand-coded counterpoint rules running on the ILLIAC I at the University of Illinois (Hiller & Isaacson, 1959). Subsequent algorithmic-composition work in the lineages of Cope (Cope, 1996), Ebcioglu (Ebcioglu, 1988), and Pachet (Pachet, 2003) developed increasingly sophisticated explicit-operator systems grounded in style-specific rule sets. Neural sequence modeling for symbolic music began in earnest with the Music Transformer (Huang et al., 2019) and continued through Museformer (Yu et al., 2022), FIGARO (von Rütte et al., 2023), MuseCoco (Lu et al., 2023), and the 2025 wave of personalizable infilling models (MIDI-GPT, MIDI-RWKV (Zhou-Zheng & Pasquier, 2025), Calliope). On the audio side, conditional neural generation expanded from short-clip generation (Jukebox (Dhariwal et al., 2020), MusicLM, MusicGen) to long-form lyrics-to-song systems (Suno, Udio, YuE, Muse, DiffRhythm, ACE-Step, LeVo). Voice cloning developed in parallel through ElevenLabs, Cartesia Sonic, and MiniMax Speech 2.5.

The MIREX 2025 Symbolic Music Generation challenge formalizes the task closest to historical completion: given a four-measure piano prompt, generate a musically coherent twelve-measure continuation. None of the competing systems situate themselves explicitly within the centuries-long musicological tradition of the previous subsection, although they are, structurally, addressing the same problem.

2.3. What Is Missing

The two literatures rarely refer to each other. Musicology has accumulated a deep understanding of what makes a continuation stylistically valid, expressed in categories specific to musical style: voice leading, motivic development, formal proportion, harmonic syntax. Computational work has produced sophisticated systems that perform completion at scale, evaluated in categories specific to ML: audio fidelity, distributional similarity, downstream user preference. Neither has produced a formalism through which the other’s insights become technically actionable. The framework of Section 3 is intended to bridge this gap, not as the only possible bridge, but as one that respects both traditions: musicology’s insistence on style as a structured, multi-feature object, and ML’s expectation that any framework be operationalizable on real outputs.

3. Composition-Invariant Operators for Stylistic Completion

3.1. Setup

Let $\mathcal{A}$ be a musical action space: the set of elementary compositional moves (motif transformations, voice-leading steps, harmonic progressions, orchestration decisions). A musical trajectory is a sequence in $\mathcal{A}^*$. A *completer* is an

operator $C : \mathcal{A}^* \rightarrow \mathcal{A}^*$ that extends a partial trajectory.

A composer O ’s style induces a measure μ_O over trajectories, intuitively the distribution of sequences that O would produce. The support of μ_O defines the space of “stylistically legitimate” trajectories for O .

3.2. Composition-Invariance

A property P of trajectories is **composition-invariant** under μ_O if and only if for any operator C in the support of μ_O (i.e., any completer that produces stylistically legitimate continuations), P is preserved under the action of C .

Equivalently, P is composition-invariant if its variance under the family of legitimate completers $\mathcal{C}_O := \{C : C \sim \mu_O\}$ is zero. Composition-invariant properties lie in the *kernel* of the style.

Exact invariance is rare in practice; we therefore work with the ϵ -relaxation, in which P ’s expected deviation under $C \sim \mathcal{C}_O$ is bounded by some tolerance ϵ , yielding a relaxed kernel $\mathcal{K}_\epsilon(O)$. The case study in Section 5 uses this relaxed form, with ϵ calibrated empirically.

3.3. Kernel vs. Cloud

This definition partitions stylistic features into two classes:

- **Kernel:** Properties that are composition-invariant. These are identity-constituting: violating them produces a continuation that is not recognizably “in the style of O .” Süssmayr could not have written atonal continuations and remained Mozartean.
- **Cloud:** Properties that vary across legitimate completers. These represent stylistic choices: Süssmayr could have orchestrated the *Lacrimosa* differently and still produced a Mozartean completion.

The distinction has structural similarity to questions copyright doctrine has struggled to formalize, though we do not claim the kernel resolves them (see Section 6).

3.4. Interface with Musicological Theory

Several existing musicological frameworks are compatible with the kernel concept:

- **Schenkerian analysis** (Schenker, 1979): The *Ursatz* and prolongational structure are candidates for kernel properties, deep structural features that any tonal completion must preserve.
- **Lerdahl & Jackendoff** (Lerdahl & Jackendoff, 1983): Grouping, metrical, time-span, and prolongational reductions provide hierarchical invariants.

- **Lewin’s transformational theory** (Lewin, 1987): Musical actions as group elements; the group structure itself may be a kernel property.
- **Tymoczko’s voice-leading geometry** (Tymoczko, 2011): Parsimony bounds (step-size distributions between adjacent chord tones) provide quantitative kernel candidates.
- **Kramer’s musical time** (Kramer, 1988): Temporal strategies and listening modes as style-level invariants.

We do not claim these frameworks *are* the kernel; we claim they are compatible with the kernel concept and provide musicologically grounded diagnostics.

This separation is methodologically important: candidate kernel features are drawn from theory independent of any particular completer, allowing kernel-membership to be tested empirically rather than presupposed.

4. AI Completers as Generalized Süssmayrs

4.1. A Taxonomy by Operator Visibility

The trajectory from Illiac Suite to Suno can be read along a single axis: the visibility of the operators acting on the musical action space. We distinguish two regimes.

Explicit-operator era (1956 to roughly the late 2010s).

The operators of the completer are written down. Hiller and Isaacson’s *Illiac Suite* used hand-coded sixteenth-century counterpoint rules (Hiller & Isaacson, 1959); subsequent algorithmic-composition work in the Cope (Cope, 1996), Ebcioglu (Ebcioglu, 1988), and Pachet (Pachet, 2003) lineages encoded harmonic templates, stylistic constraints, and grammar-based generation explicitly; recent work continues this tradition in domains such as game sonification (Zheng, 2024). Composition-invariance is, in this regime, mechanically checkable: a property is in $\mathcal{K}(O)$ iff it is preserved by the rule set, and the rule set is inspectable.

Implicit-operator era (roughly late 2010s to present).

The operators are learned. The Music Transformer (Huang et al., 2019) introduced large-scale neural sequence modeling for symbolic music; the lineage continued through Museformer (Yu et al., 2022), FIGARO (von Rütte et al., 2023), MuseCoco (Lu et al., 2023), and the 2025 wave including MIDI-GPT and MIDI-RWKV (Zhou-Zheng & Pasquier, 2025). On the audio side, the same shift unfolded in parallel: Jukebox (Dhariwal et al., 2020), MusicLM (Agostinelli et al., 2023), MusicGen (Copet et al., 2023), then YuE (Yuan et al., 2025), Muse (Jiang et al., 2026), DiffRhythm, ACE-Step, LeVo, Suno, and Udio. Voice followed the same trajectory, from concatenative and parametric synthesis to ElevenLabs, MiniMax Speech 2.5, and Cartesia Sonic.

The transition is consequential for the composition-invariance question. In the explicit era, kernel membership was a property of an inspectable rule book. In the implicit era, operators are encoded in network weights and inference-time procedures; kernel membership can no longer be read off and must be diagnosed empirically. The diagnostic battery of Section 5 is one response to this fact.

4.2. Stylistic Completion Across Modalities

The operator framework is modality-agnostic. The same task structure (given a partial trace and a stylistic prior, extend the trace under the prior) appears in three distinct contemporary AI domains, each with its own active research community largely unaware of the others’ work.

Symbolic music. The 2025 wave of symbolic music infilling models is, structurally, the Süssmayr task formalized as an ML benchmark. The MIREX 2025 Symbolic Music Generation challenge defines its task as: given a four-measure piano prompt and an optional pickup measure, generate a musically coherent twelve-measure continuation. MIDI-GPT introduced a Bar-Fill representation in which masked bars are replaced with `Infill_Bar` tokens and target content is moved to a sequence-end region; MIDI-RWKV (Zhou-Zheng & Pasquier, 2025) adapts this for personalizable, long-context, multi-track symbolic infilling, with state-tuning for per-user adaptation. Calliope productionizes infilling for compositional workflows. None of these systems invoke the historical lineage of stylistic completion. They present themselves as different solutions to a shared task that the framework of Section 3 names: continue a partial musical trace under a prior.

Audio music. Long-form lyrics-to-song systems (Suno, Udio, YuE, DiffRhythm, ACE-Step, LeVo, and Muse) differ in tokenization, training data, and conditioning but share the task structure with the symbolic systems above. Muse is particularly instructive: it trains on 116k synthetic songs generated by Suno V5 and demonstrates that a Qwen-based LM extended with discrete audio tokens can match commercial systems on text–music similarity and aesthetic quality at modest scale. The use of synthetic training data from another generator is, on our framework, a notable choice; we discuss its kernel-level consequences below.

Voice. Voice cloning is stylistic completion at the utterance level. Given a short audio sample of a speaker (five seconds suffices for MiniMax Speech 2.5), the system produces continuations of arbitrary text in that speaker’s voice. Cartesia’s Sonic is architecturally relevant beyond its task domain: it is built on the state-space-model lineage of S4 (Gu et al., 2022) and Mamba (Gu & Dao, 2024), in which signals are processed by parameterized continuous-time

linear operators. The architecture instantiates explicit operators on a continuous signal space, a literal realization of the operator-substrate framing, and provides one of the cleanest current examples of learned operators that retain inspectable structure.

Cross-modal product companies make the unity especially visible. MiniMax ships Speech 2.5 (voice cloning) and music-1.5 (music generation) within a single product platform; the framework treats these as instances of one problem class differing in tokenization, not in structure.

4.3. What the Framework Predicts About Current Systems

Three predictions follow from treating $\mathcal{K}(O)$ as the central object of stylistic completion.

Personalization works because it fits per-user kernels.

Tools that adapt to a small per-user corpus (MIDI-RWKV’s state-tuning, voice cloning from a five-second sample) are, on this framework, explicitly fitting a kernel to a single individual. Their effectiveness despite small data scales is consistent with the framework’s prediction that kernel structure is concentrated in a small number of features rather than distributed across the full support of the distribution. Models that target central-tendency style without per-user fitting necessarily produce more frequent kernel violations relative to any specific composer; the homogeneity of generic AI music outputs is the predicted symptom.

Long-form coherence is hard because composition-invariance compounds. A valid continuation must remain inside $\mathcal{C}_O$ at every step; over a long horizon, learned operators tend to drift toward training-distribution central tendency rather than tracking a specific composer’s kernel. The persistent difficulty of long-form coherence noted by the YuE authors and across the long-form-song literature is consistent with this: long-horizon kernel preservation requires the operator to remain inside $\mathcal{C}_O$ across many compositional decisions, while training objectives (next-token prediction, classifier-free guidance) reward central-tendency rather than kernel-faithful behavior.

Synthetic-data training risks kernel collapse. When a generation system is trained on outputs of another generation system (as Muse trains on Suno V5), the operator population is filtered through the source system’s kernel. Iterated, this drives the trainee’s kernel toward the central tendency of its source, narrowing the space of supported stylistic identities. The framework predicts that synthetic-data pipelines, while effective in the short term, accelerate homogenization. The connection to model-collapse arguments in language modeling (Shumailov et al., 2024) is direct: kernel collapse is the music-specific instance of a

general phenomenon.

These predictions are testable. We do not test them here (Section 5’s case study is illustrative rather than predictive), but we view them as concrete commitments of the framework that future empirical work could verify or refute.

5. A Diagnostic Battery

5.1. Six Candidate Invariants

We propose a battery of musicologically-grounded composition-invariants, each a candidate kernel feature for a wide class of composers in the Western tonal canon. The battery is not exhaustive; it is a starting set chosen for two properties: each has a defensible musicological provenance, and each is computable on outputs of contemporary symbolic and audio music systems. Table 1 summarizes.

Feature ϕ	Domain	Provenance
Voice-leading step distribution	sym/audio	Tymoczko (2011); contrapuntal pedagogy
Motivic interval n -gram sets	symbolic	Lerdahl & Jackendoff (1983); Réti (1951)
Harmonic-rhythm regularity	sym/audio	Cone (1968); style-period analysis
Phrase-length distribution	sym/audio	Caplin (1998); period theory
Cadential vocabulary	symbolic	Schmalfeldt (2011); cadence theory
Orchestration doubling	sym/audio	Berlioz (1843); orchestration manuals

Table 1. Candidate composition-invariant diagnostics with musicological provenance.

For each, the *kernel hypothesis* is that the feature lies within $\mathcal{K}_\epsilon(O)$ for some ϵ small enough to discriminate in-style from out-of-style continuations. This is a hypothesis tested per composer, not asserted globally. We compute features 1 and 2 below and apply them to a case study.

5.2. Computation

Voice-leading step distribution. Given a symbolic input, we extract vertical sonorities at each onset, sort the pitch set at each onset, and pair pitches across adjacent sonorities by minimum displacement. The resulting collection of step sizes (in semitones) is summarized by its mean $\bar{s}$ and the proportion of steps ≤ 2 , which we denote ρ_2 . For a composer whose practice involves tight voice leading, both statistics are kernel candidates: a continuation with substantially larger $\bar{s}$ or substantially smaller ρ_2 has departed from a feature any in-style continuation would preserve.

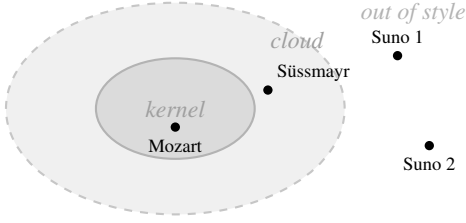


Figure 1. Composition-invariance partitions stylistic features into kernel, cloud, and out-of-style regions. Süssmayr falls inside the cloud (passing the voice-leading kernel candidate, failing the motivic one); two Suno V5 runs fall outside any Mozartean band, and far apart from each other. Illustrative; values in Table 2.

Motivic interval n -gram set. Given a symbolic input, we extract the top-voice pitch sequence, convert to consecutive interval differences (in semitones), and count n -grams ($n = 3$ in our case study). The motivic-overlap statistic between two traces is the multiset Jaccard:

$$J(c_1, c_2) = \frac{\sum_k \min(c_1[k], c_2[k])}{\sum_k \max(c_1[k], c_2[k])}, \quad (1)$$

where c_1, c_2 are the interval- n -gram counters. The kernel hypothesis is positional: continuations that draw their motivic material from the prompt share interval n -grams with it; continuations that introduce uncorrelated material do not.

5.3. Case Study: The *Lacrimosa*

We apply the two diagnostics to the *Lacrimosa* of Mozart’s *Requiem*, a well-controlled case because Mozart’s autograph terminates at measure 8, with multiple human completions in the literature. We compute each diagnostic on three traces:

- **Prompt (Mozart).** Measures 1–8, from Mozart’s autograph.
- **Süssmayr completion.** Measures 9–30, from the published Süssmayr score.
- **Suno V5 generation.** Two independent audio continuations prompted in the *Lacrimosa* style, transcribed via Basic Pitch.

Trace	$\bar{s}$	ρ_2	J
Prompt (Mozart, mm. 1–8)	4.07	0.40	—
Süssmayr (mm. 9–30)	3.64	0.50	0.055
Suno V5 ($n=2$ runs) [†]	7.1–13.7	0.26–0.31	0.000

Table 2. Voice-leading and motivic diagnostics for the *Lacrimosa*. Suno values are reported as the range across two independent generations of the same prompt. [†]Suno output transcribed via Basic Pitch.

The Süssmayr completion’s voice-leading statistics ($\bar{s} = 3.64$, $\rho_2 = 0.50$) fall close to the Mozart prompt’s ($\bar{s} = 4.07$, $\rho_2 = 0.40$), with Süssmayr in fact registering tighter ($\bar{s}$ 11% smaller) and more stepwise (ρ_2 25% larger). Both differences fall well inside the within-composer baseline established on K.545, where Mozart-with-Mozart already exhibits a 31% shift in $\bar{s}$ across consecutive eight-bar windows; Süssmayr’s deviation from the *Lacrimosa* prompt is, by this measure, smaller than Mozart’s deviation from himself. The result is consistent with the historical consensus that Süssmayr’s command of part-writing was idiomatic; the slight tightening of his voice-leading relative to the prompt is also consistent with a recurring musicological observation that Süssmayr’s continuations smooth Mozart’s rougher edges into more textbook part-writing.

The motivic-overlap statistic, $J = 0.055$, is by contrast low. Süssmayr’s continuation does not draw heavily from the specific interval n -grams of the prompt’s opening eight measures: of the material he composes after Mozart’s autograph terminates, he chooses not to derive his figures from systematic transformation of the opening’s motivic stock, but to introduce new material appropriate to the text and harmonic situation. As a robustness check, we computed motivic overlap using skip-1 trigrams (allowing one intervening interval); results are qualitatively similar ($J = 0.037$), reinforcing that Süssmayr’s motivic vocabulary diverges from the prompt.

The two diagnostics measure distinct kernel candidates. By the voice-leading candidate, Süssmayr’s continuation is well inside the tolerance of “Mozartean”; by the motivic candidate, it is not.

The Suno V5 continuations present a starkly different profile. Across two independent generations, $\bar{s}$ ranged from 7.05 to 13.67 and ρ_2 from 0.26 to 0.31, with $J = 0.000$ in both cases. The lower value of $\bar{s}$ alone already exceeds the within-Mozart baseline (31%) by more than a factor of two; the higher value enters territory inconsistent with stepwise voice-leading at all. The variance *between* the two Suno generations is also notable: the two runs differ from each other in $\bar{s}$ by 94%, while Süssmayr’s human continuation deviates from the Mozart prompt by only 11%. AI generation appears, at this preliminary scale, to exhibit substantially higher cross-sample variance in voice-leading statistics than is observed across competent human stylistic completers; each Suno output sits firmly outside the within-Mozart band, but the outputs are also far from each other. The motivic overlap $J = 0.000$ is robust across both generations: not a single consecutive interval 3-gram from Mozart’s prompt appears in either Suno output.

The Suno V5 web interface does not expose temperature, top- p , or guidance scale to end users; the variance we report reflects sampling stochasticity at the platform’s default

settings, which is what typical users encounter.

Variability between Suno runs partly reflects audio-generation stochasticity and partly reflects sensitivity of voice-leading metrics to transcribed sonority density: sparser Basic Pitch output yields larger $\bar{s}$ values. The qualitative finding, that Suno V5 continuations sit outside the within-Mozart kernel band on every observed run, is robust to this confound; absolute values should be interpreted with the transcription pipeline in mind.

Where Süssmayr passes the voice-leading kernel and fails the motivic kernel, the AI fails both (Figure 1). This three-way comparison illustrates the framework’s discriminative power: a human completer trained in eighteenth-century counterpoint preserves voice-leading parsimony even when introducing new motivic material, while a contemporary neural generator produces completions that depart from both kernel candidates.

Calibration: within-composer variation. To calibrate within-composer variation with like-texture material, we run the same diagnostics on Mozart’s Kyrie from the same Requiem (K.626), comparing mm. 1–8 to mm. 9–16. The Kyrie is entirely Mozart’s autograph and shares the SATB-plus-orchestra texture of the *Lacrimosa*. This within-Mozart comparison yields $\bar{s}$ shift $5.06 \rightarrow 4.31$ (15% decrease) and ρ_2 shift $0.38 \rightarrow 0.37$ (essentially unchanged). Süssmayr’s *Lacrimosa* deviation from the Mozart prompt ($\bar{s}$ 11%, ρ_2 25%) is comparable to or smaller than this within-Mozart baseline for $\bar{s}$, supporting the claim that his voice-leading lies inside the Mozartean kernel. The motivic overlap within the Kyrie ($J = 0.167$) is notably higher than Süssmayr’s overlap with the *Lacrimosa* prompt ($J = 0.055$), consistent with the observation that Süssmayr introduced new motivic material rather than developing the opening’s stock.

The Kyrie is a double fugue with more active voice-leading than the homophonic *Lacrimosa*; absolute $\bar{s}$ values are accordingly higher. The comparison nonetheless establishes a same-texture, same-work baseline that the K.545 solo-piano comparison cannot provide.

5.4. Scope and Limits

The case study is illustrative. It establishes that the diagnostics are computable on real material, that they discriminate, and that distinct candidates expose distinct facets of kernel structure. It does not establish that voice-leading parsimony and motivic n -gram overlap fully specify $\mathcal{K}(\text{Mozart})$, nor that any single diagnostic is decisive. Establishing kernel membership at scale requires replication across many prompts and many would-be completers, human and machine, and is beyond the scope of this paper. We intend this as an existence proof: the framework of Section 3 admits a tractable empirical realization.

A natural controlled experiment would render a known symbolic completion (e.g., the Süssmayr score) through an acoustic synthesizer, pass it through the same Basic Pitch pipeline, and recompute the diagnostics; the resulting transcription-induced shift would isolate pipeline noise from generative drift. We view this as a clear immediate next step.

Implementation details are provided in Section A.

6. Discussion

6.1. Beyond Audio-Quality Evaluation

Evaluation of generative music systems is dominated by audio-quality and distributional similarity metrics: Fréchet Audio Distance (Kilgour et al., 2019), CLAP-based text-music similarity, phoneme error rates for sung lyrics. These are valuable but blind to the structural question musicology has spent a century articulating: whether a generation reproduces what makes a composer’s voice their own. Composition-invariants supply targets that are musicologically meaningful and computable on the same outputs current pipelines already produce. The diagnostic battery of Section 5 is a starting point rather than a fixed taxonomy; the framework permits new feature maps to be added as their kernel-membership is empirically established.

6.2. Licensing, Consent, and the Limits of Legal Vocabulary

The 2026 settlements (UMG/Udio in October 2025, Warner/Suno in November 2025) anticipate licensed AI music services launching in 2026. These services will require operational definitions of what an artist’s catalog “permits” the AI to generate. Current licensing language is largely silent on the question, defaulting to whole-catalog or whole-artist permission grants without per-feature granularity.

We make no claim that the stylistic kernel $\mathcal{K}(O)$ is a legal definition of protectable style. Under U.S. copyright doctrine, style itself is not copyrightable; the boundary between unprotected style and protected expression has long been contested in case law (Henderson et al., 2023) and is not the framework’s to settle. Our claim is more modest: licensing discussions, artist-consent flows, and platform-internal generation policies all require *some* operational definition of “what features make a continuation belong to this artist,” and they currently lack one grounded in either musicology or measurement. The kernel framework supplies a candidate technical vocabulary that is independent of any particular legal regime. Whether and how it is taken up by legal practice is a separate question we do not address here.

6.3. Co-Creative Tools and the Homogenization Risk

A widely noted concern about generative AI in creative domains is idea homogenization: tools trained on the central tendency of a corpus tend to produce outputs that converge on that tendency, narrowing the diversity of work users produce (Doshi & Hauser, 2024). The composition-invariance framework recasts a central design question for human-AI co-creative tools: *which features of a user’s evolving stylistic identity should the tool preserve, and which should it actively expand?* A tool that holds kernel features fixed while diversifying cloud features is, in this language, a tool that supports rather than flattens its user’s voice.

The kernel/non-kernel split offers a principled angle on the homogenization problem. A co-creative tool that preserves user-specific kernel features while expanding the space of non-kernel choices is, by construction, supporting the user’s stylistic identity while widening their interpretive options. Recent personalizable infilling work such as MIDI-RWKV (Zhou-Zheng & Pasquier, 2025) gestures at this; its state-tuning protocol can be read as fitting a small-scale per-user kernel, but no current system separates the two concerns explicitly. We see this as a concrete design direction for the next generation of co-creative tools.

6.4. The Voice Modality and a Different Legal Landscape

Voice differs from music in legal regime: voice and likeness are increasingly protected under right-of-publicity statutes and dedicated legislation (e.g., Tennessee’s ELVIS Act, 2024) rather than under copyright. For voice cloning systems such as Cartesia Sonic and MiniMax Speech 2.5, “what makes a voice this person’s voice” is not only a technical or licensing question but a statutory one. We do not pursue this thread here. We only note that the kernel framework, being modality-agnostic, extends naturally to voice, and that its outputs may map more directly onto the legal questions in the voice domain than in the music-copyright domain.

6.5. Limitations

Four limitations are worth flagging. First, $\mathcal{K}(O)$ is empirically inferred, not given; the kernel members we propose in Section 3 and Section 5 are candidates, and kernel-membership for each is itself a hypothesis subject to test. Second, the framework assumes a stable composer prior, but compositional style evolves across a career (Beethoven’s late kernel differs from his early one), and we do not yet handle this temporal structure. Third, our case studies are drawn from the Western tonal canon; cross-cultural application requires both new feature maps and revised assumptions about hierarchical structure. Fourth, a multi-completer comparison (Eybler, Maunder, Levin, Druce) would substantially

strengthen the empirical analysis; symbolic encodings of the modern recompletions are not publicly available, and we leave this to future work.

7. Conclusion

Stylistic completion has structured Western music for centuries: from Süssmayr’s continuation of an unfinished *Requiem*, through Hiller and Isaacson’s hand-coded operators in the Illiac Suite, to the learned operators of contemporary symbolic, audio, and voice systems. The transition from explicit to implicit operators leaves a question musicology has refined for over a century but ML has not yet inherited: *which features of an original style must any continuation preserve?*

Our proposed answer is composition-invariance. Stylistic completers are operators on a musical action space; the kernel of composition-invariants under a composer’s prior identifies the features distinguishing stylistic identity from interpretive choice. The framework is modality- and architecture-independent; it accommodates Bach’s contrapuntal explicitness alongside Suno’s opacity.

Three directions seem most productive: empirical kernel discovery at scale across many composers and completers; temporal kernels that handle stylistic evolution within a career; and cross-cultural extension to traditions where the assumptions of hierarchical pitch organization and harmonic syntax do not apply.

Süssmayr worked from sketches and the memory of his teacher’s voice. The systems shipping in 2026 work from training corpora and learned conditioning. Both attempt the same task. The framework we propose is a candidate vocabulary for asking, of either, *whose voice does this continuation belong to?*

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A. Implementation

We describe each diagnostic computation at the algorithmic level. The implementation uses `music21` for symbolic parsing and `numpy` for numerical operations.

Sonority extraction. A score is parsed into a flat list of (onset, MIDI pitch) events; `Note` objects contribute one event each, `Chord` objects contribute one event per constituent pitch at the chord’s onset. Events are sorted by onset. A sliding window groups events into sonorities: events whose onset is within w quarter notes of the current sonority’s start are added to that sonority; when an event falls outside the window, the current sonority is closed (emitted as a sorted pitch list) and a new sonority begins. We use $w = 1.0$ quarter note throughout, which absorbs the slight onset offsets typical of orchestral MIDI where parts attack on adjacent subdivisions. The procedure returns a temporally ordered list of pitch sets.

Voice-leading steps and statistics. Each adjacent pair of sonorities is paired by sorted position: lowest with lowest, second-lowest with second-lowest, and so on, truncating to the smaller cardinality. The absolute semitone difference of each pair is collected across all adjacent pairs. Summary statistics are the mean ($\bar{s}$) and the proportion not exceeding 2 semitones (ρ_2). The sorted-position heuristic is sufficient for illustrative diagnostics; for formal use, Hungarian-algorithm optimal assignment would be preferable.

Top voice and motivic intervals. The top voice is the sequence of maximum pitches across sonorities. Its consecutive signed-semitone differences form the motivic interval sequence, which is transposition-invariant by construction. Rhythmic normalization is not applied; intervals are pitch-only, not duration-conditioned. We use $n = 3$ in main results.

Standard and skip n -grams. For an interval sequence of length L , the standard n -gram counter records every contiguous length- n tuple. The skip- k variant uses positions spaced $k+1$ apart, recording tuples $(s_i, s_{i+(k+1)}, s_{i+2(k+1)}, \dots)$ for each starting index i where the full tuple fits within L .

Multiset Jaccard. For two counters c_1, c_2 over a shared domain, the multiset Jaccard is $\sum_k \min(c_1[k], c_2[k]) / \sum_k \max(c_1[k], c_2[k])$, summed over the union of keys, with the convention $J = 0$ when both counters are empty.

Bootstrap confidence intervals. A statistic f is computed on each of $B = 1000$ resamples drawn with replacement at the original length. The 95% CI is the $(2.5, 97.5)$ per-

centile range. The random number generator is seeded for reproducibility.